

ZHAW SCHOOL OF ENGINEERING
INSTITUTE OF MECHATRONIC SYSTEMS

Betaflight – Offline Controller Tuning

Data-Driven Tuning of Rate and Altitude Controllers for FPV Drones

Bachelor Thesis

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March 4, 2026

Declaration of Originality

We hereby declare that this thesis is our own work and has been composed solely by the undersigned. We have not used any sources or aids other than those indicated. All passages taken verbatim or in substance from published or unpublished works of others have been clearly marked as such. This thesis has not been submitted, in whole or in part, for any other degree or examination at this or any other institution.

Winterthur, March 4, 2026

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Zusammenfassung

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Abstract

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Preface

This thesis was carried out during the spring semester 2026 at the Institute of Mechatronic Systems (IMS) at the ZHAW School of Engineering. It continues the work started in the preceding project thesis conducted in the fall semester 2025.

We would like to thank our supervisor, Michael Peter, for...

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1. Introduction

1.1. Background and Motivation

1.2. Research Questions and Objectives

1.3. Document Structure

This thesis is organized as follows. Chapter 2 provides the theoretical foundations. . . Chapter 3 describes the methodology. . . Chapter 4 details the implementation. . . Chapter 5 presents the experimental results. . . Chapter 6 discusses the findings and outlines future work.

2. Theoretical Foundations

2.1. Betaflight Control Architecture

2.2. System Identification via Chirp Excitation

2.3. Frequency Response Estimation

2.4. Closed-Loop Analysis

2.5. Cascaded Control Structures

3. Methodology

3.1. Tuning Workflow Overview

3.2. Flight Test Protocol

Hello, this is a placeholder for the flight test protocol section. Here we will describe the specific maneuvers and conditions under which the flight tests will be conducted to collect data for system identification and controller tuning.

3.3. Signal Processing Pipeline

3.4. Controller Design Procedure

3.5. Altitude Hold System Identification

4. Implementation

4.1. Software Architecture

4.2. MATLAB Implementation

4.3. Python Implementation

4.4. Betaflight Firmware Modifications

4.5. Validation Framework

5. Results

5.1. Rate Controller Tuning Results

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5.5. Flight Test Validation

6. Discussion and Outlook

6.1. Discussion of Results

6.2. Limitations

6.3. Future Work

6.4. Conclusion

Declaration of Generative AI Use

The following generative AI systems were used during the preparation of this thesis. Their use was limited to the functions described below. All generated content was critically reviewed and verified by the authors.

A. Project Management

A.1. Project Assignment

A.2. Project Schedule

B. Additional Figures and Data